

# MIXED VOLUMES OF OKOUNKOV BODIES

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ABSTRACT. Let  $X$  be a connected smooth projective variety of dimension  $n$ . We prove that the movable product of any big line bundles  $L_1, \dots, L_n$  equals  $n!$  times the supremum of the mixed volumes of their Okounkov bodies over all valuations.

## 1. INTRODUCTION

Let  $X$  be a connected smooth projective variety of dimension  $n$ . For a big line bundle  $L$  and a surjective valuation

$$\nu: \mathbb{C}(X)^\times \longrightarrow \mathbb{Z}_{\text{lex}}^n$$

that is trivial on  $\mathbb{C}^\times$ , write  $\Delta_\nu(L) \subseteq \mathbb{R}^n$  for the associated Okounkov body. Its Euclidean volume recovers the volume, or equivalently the top movable self-intersection, of  $L$ :

$$(1.1) \quad n! \operatorname{vol} \Delta_\nu(L) = \operatorname{vol}(L) = \langle L^n \rangle.$$

For the full valuation class used here, Okounkov bodies depend only on numerical classes [Xia26a, Proposition 10.3.2]; the flag-valuation case is [LM09, Proposition 4.1]. Conversely, Jow proved that two big divisors on a normal complex projective variety are numerically equivalent if their Okounkov bodies agree for every admissible flag [Jow10, Theorem A]. Thus the collection of all Okounkov bodies is a faithful numerical invariant.

The first conclusion of this paper is that this collection also recovers mixed movable products.

**Theorem 1.1.** *Let  $L_1, \dots, L_n$  be big line bundles on  $X$ , then*

$$(1.2) \quad \langle L_1, \dots, L_n \rangle = n! \sup_\nu \operatorname{vol}(\Delta_\nu(L_1), \dots, \Delta_\nu(L_n)),$$

where  $\nu$  ranges over the valuations above.

Here  $\operatorname{vol}$  denotes mixed volume, normalized by  $\operatorname{vol}(K, \dots, K) = \operatorname{vol}(K)$ . For ample line bundles, Wilms proved that a suitable single admissible flag realizes the mixed intersection number [Wil25, Theorem 1.1].

We now pass to its singular-metric refinement. A Hermitian big line bundle  $(L, h)$  will mean a big line bundle endowed with a singular psh metric with positive volume (in the sense of Cao). In [Xia25], the author introduced the partial Okounkov body  $\Delta_\nu(L, h) \subseteq \mathbb{R}^n$  extending the usual Okounkov body construction. The volume theorem [Xia25, Theorem A] gives

$$(1.3) \quad n! \operatorname{vol} \Delta_\nu(L, h) = \operatorname{vol} \operatorname{dd}^c h,$$

where the volume on the right-hand side is taken in the sense of Cao [Cao14]. Moreover, the family of partial Okounkov bodies over all rank- $n$  valuations determines the  $\mathcal{I}$ -singularity type of  $h$  [Xia25, Theorem B], a metric analogue of Jow's theorem.

A formula of this type was first proposed in [Xia25, Conjecture 8.8]. We prove it here.

**Theorem 1.2.** *Let  $X$  be a connected smooth projective variety of dimension  $n$ . For  $1 \leq i \leq n$ , let  $L_i$  be a big line bundle and let  $h_i$  be a psh metric on  $L_i$  whose volume is positive. Then*

$$(1.4) \quad \operatorname{vol}(\operatorname{dd}^c h_1, \dots, \operatorname{dd}^c h_n) = n! \sup_\nu \operatorname{vol}(\Delta_\nu(L_1, h_1), \dots, \Delta_\nu(L_n, h_n)).$$

Here  $\nu$  runs over all valuations as above.

Again, the mixed volume is taken in the sense of Cao; by [Xia26b, Definition 3.2, Proposition 3.3, and Theorem 3.5], it coincides with the mixed volume in the sense of Darvas–Xia recalled in Section 2. Taking each  $h_i$  to have minimal singularities reduces Theorem 1.2 to Theorem 1.1, while its diagonal specialization is (1.3).

The proof of the  $\leq$  direction in (1.4) can be easily reduced to Wilms' theorem [Wil25, Theorem 1.1]. Conversely, the  $\geq$  direction can be reduced to the  $\geq$  direction in (1.2). We shall apply a suitable toric degeneration to further reduce to the toric case, where the results are essentially known.

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## 2. PRELIMINARIES

We follow the language and notation of [Xia26a]. All products of closed positive currents are non-pluripolar products. We use  $dd^c = (2\pi)^{-1}i\partial\bar{\partial}$ , and  $\mathbb{N} = \{0, 1, 2, \dots\}$ . The notation  $\mathbb{Z}_{\text{lex}}^n$  means  $\mathbb{Z}^n$  with the lexicographic order, and  $e_1, \dots, e_n$  denotes its standard basis. A superscript  $\times$  means that zero is omitted, and  $\text{Conv } A$  denotes the convex hull of a subset  $A \subseteq \mathbb{R}^n$ . We use  $\mathbb{P}(V)$  for the space of lines in a vector space  $V$ .

**2.1. Singularities and partial Okounkov bodies.** Fix a smooth Hermitian metric  $h^0$  on a big line bundle  $L$  and write

$$\theta := c_1(L, h^0), \quad h = h^0 e^{-\varphi}, \quad \theta_\varphi := \theta + dd^c \varphi,$$

where  $\text{PSH}(X, \theta)$  is the set  $\theta$ -psh functions, and

$$\text{PSH}(X, \theta)_{>0} := \left\{ u \in \text{PSH}(X, \theta) : \int_X \theta_u^n > 0 \right\}.$$

For  $u, v \in \text{PSH}(X, \theta)$ , write  $u \preceq v$  if  $u \leq v + O(1)$ . The model envelope is

$$P_\theta[u] := \sup^* \{ v \in \text{PSH}(X, \theta) : v \leq 0, v \preceq u \},$$

where  $\sup^*$  denotes the upper-semicontinuous regularization of the pointwise supremum. A positive-mass potential  $u$  is a model potential if  $P_\theta[u] = u$ . Two positive-mass potentials are  $P$ -equivalent if their model envelopes agree. They are  $\mathcal{I}$ -equivalent if  $\mathcal{I}(\lambda u) = \mathcal{I}(\lambda v)$  for every  $\lambda > 0$ , where  $\mathcal{I}(\lambda u)$  is the multiplier ideal sheaf. The  $\mathcal{I}$ -model envelope is

$$P_\theta[u]_{\mathcal{I}} := \sup^* \{ v \in \text{PSH}(X, \theta) : v \leq 0, v \text{ is } \mathcal{I}\text{-equivalent to } u \}.$$

For an arbitrary  $\varphi \in \text{PSH}(X, \theta)$ , its volume in the sense of Cao is

$$\text{vol } \theta_\varphi := \int_X \theta_{P_\theta[\varphi]_{\mathcal{I}}}^n.$$

A potential  $\varphi \in \text{PSH}(X, \theta)_{>0}$  is called  $\mathcal{I}$ -good if

$$P_\theta[\varphi] = P_\theta[\varphi]_{\mathcal{I}}.$$

More generally, if  $L_1, \dots, L_n$  are big line bundles with  $\theta_i$  being a closed smooth real  $(1, 1)$ -form in  $c_1(L_i)$  and  $\varphi_i \in \text{PSH}(X, \theta_i)$  satisfying  $\text{vol}(\theta_i + dd^c \varphi_i) > 0$ , we write

$$\text{vol}(\theta_1 + dd^c \varphi_1, \dots, \theta_n + dd^c \varphi_n) = \int_X (\theta_1 + dd^c P_{\theta_1}[\varphi_1]_{\mathcal{I}}) \wedge \dots \wedge (\theta_n + dd^c P_{\theta_n}[\varphi_n]_{\mathcal{I}}).$$

The mixed volume defined here coincides with that defined by Cao [Xia26b, Definition 3.2, Proposition 3.3, and Theorem 3.5].

By [Xia26a, Theorem 7.1.1],  $\mathcal{I}$ -goodness is equivalent both to the mass–volume identity

$$\int_X \theta_\varphi^n = \text{vol } \theta_\varphi$$

and to the existence of potentials  $\varphi_j$  with analytic singularities such that

$$\varphi_j \xrightarrow{d_S} \varphi.$$

Here  $d_S$  is the pseudometric on singularity types introduced in [DDNL21], see also [Xia26a, Definition 6.2.1]. The approximants may be chosen so that every  $\theta_{\varphi_j}$  is a Kähler current. Following [Xia26a, Definition 1.6.1], a potential has analytic singularities if locally it is of the form

$$c \log(|f_1|^2 + \cdots + |f_N|^2) + O(1), \quad c \in \mathbb{Q}_{>0},$$

where the  $f_j$  are holomorphic and  $O(1)$  is locally bounded.

Let  $\nu: \mathbb{C}(X)^\times \rightarrow \mathbb{Z}_{\text{lex}}^n$  be surjective and trivial on  $\mathbb{C}^\times$ . Its center exists on  $X$  by properness. Choose a rational section  $\tau$  of  $L$  which is regular and nowhere vanishing near the center. For a section  $s$  of  $L^k$ , set  $\nu_\tau(s) := \nu(s/\tau^k)$ , and suppress the subscript  $\tau$  below. For every  $k \geq 1$  for which the space below is non-zero, set

$$(2.1) \quad \Delta_{\nu,k}(\theta, \varphi) := \text{Conv} \left\{ k^{-1} \nu(s) : s \in H^0(X, L^k \otimes \mathcal{I}(k\varphi))^\times \right\}.$$

Let  $\Gamma(\theta, \varphi) \subseteq \mathbb{Z}^n \times \mathbb{N}$  consist of  $(0, 0)$  and all pairs  $(\nu(s), k)$  occurring above. By [Xia26a, Theorem 10.3.1],  $\Gamma(\theta, \varphi)$  is an almost semigroup. Using the continuous Okounkov-body map  $\Delta$  of [Xia26a, Theorem 10.1.2], define  $\Delta_\nu(\theta, \varphi) = \Delta_\nu(L, h) := \Delta(\Gamma(\theta, \varphi))$ . Changing the rational trivialization translates all the bodies in (2.1); none of the mixed volumes below is affected.

We record the properties used in the proof. We write  $\varphi \preceq_{\mathcal{I}} \psi$  when  $\mathcal{I}(\lambda\varphi) \subseteq \mathcal{I}(\lambda\psi)$  for every  $\lambda > 0$ . Let  $\mathcal{K}_n$  be the space of non-empty compact convex subsets of  $\mathbb{R}^n$ , and let  $d_{\text{Haus}}$  be the Hausdorff distance on  $\mathcal{K}_n$ . We write  $\text{vol}(K)$  for the Euclidean  $n$ -volume of  $K \in \mathcal{K}_n$ . First,

$$(2.2) \quad \text{vol} \Delta_\nu(\theta, \varphi) = \frac{1}{n!} \text{vol} \theta_\varphi$$

by [Xia26a, Corollary 10.3.2]. The right-hand side is independent of  $\nu$ . Second, the map

$$\Delta_\nu(\theta, \cdot): (\text{PSH}(X, \theta)_{>0}, d_S) \longrightarrow (\mathcal{K}_n, d_{\text{Haus}})$$

is continuous by [Xia26a, Theorem 10.3.2]. Third, for another Hermitian big line bundle  $(L', h')$ , singularity comparison and tensor products give

$$(2.3) \quad \varphi \preceq_{\mathcal{I}} \psi \implies \Delta_\nu(\theta, \varphi) \subseteq \Delta_\nu(\theta, \psi),$$

$$(2.4) \quad \Delta_\nu(L, h) + \Delta_\nu(L', h') \subseteq \Delta_\nu(L \otimes L', h \otimes h'),$$

by [Xia26a, Propositions 10.3.8 and 10.3.10]. In particular, the partial body is contained in the ordinary Okounkov body  $\Delta_\nu(L)$ , obtained from the same construction using the complete linear series  $H^0(X, L^k)$ . The latter satisfies

$$(2.5) \quad \text{vol} \Delta_\nu(L) = \frac{1}{n!} \text{vol}(L)$$

by [Xia26a, Corollary 10.3.1], where  $\text{vol}(L) := \lim_{k \rightarrow \infty} n! k^{-n} h^0(X, L^k)$ .

We will also use basic facts about model envelopes. By [Xia26a, Corollary 3.1.2],  $P_\theta[\varphi]$  is a model potential. The equality  $d_S(\varphi, P_\theta[\varphi]) = 0$  follows from [Xia26a, Proposition 6.2.2], and  $P$ -equivalent potentials are  $\mathcal{I}$ -equivalent by [Xia26a, Proposition 3.2.9]. Consequently, replacing a potential by its model envelope changes neither its  $d_S$ -class nor its partial Okounkov body. If  $\varphi$  has analytic singularities, then so does  $P_\theta[\varphi]$  by [Xia26a, Proposition 3.2.10]. We will separately use the  $\mathcal{I}$ -model replacement. By [Xia26a, Proposition 3.2.6],  $\varphi$  and  $P_\theta[\varphi]_{\mathcal{I}}$  are  $\mathcal{I}$ -equivalent, so they have the same partial Okounkov bodies. If  $\text{vol} \theta_\varphi > 0$ , then

$$\int_X \theta_{P_\theta[\varphi]_{\mathcal{I}}}^n = \text{vol} \theta_\varphi > 0,$$

and  $P_\theta[\varphi]_{\mathcal{I}}$  is an  $\mathcal{I}$ -good model potential by [Xia26a, Example 7.1.2].

**Lemma 2.1.** *Let  $u_j, u \in \text{PSH}(X, \theta)_{>0}$  be model potentials such that  $u_j \xrightarrow{d_S} u$ . The sequence  $(u_j)_j$  has a further subsequence, still indexed by  $j$ , for which there are increasing model potentials  $\eta_j$  and decreasing model potentials  $\psi_j$  satisfying*

$$\eta_j \leq u_j \leq \psi_j, \quad \eta_j \leq u \leq \psi_j, \quad \eta_j \xrightarrow{d_S} u, \quad \psi_j \xrightarrow{d_S} u.$$

*Proof.* By [Xia26a, Corollary 6.2.5], the masses  $\int_X \theta_{u_j}^n$  have a positive lower bound after finitely many terms. So we can apply [Xia26a, Proposition 6.2.3] and then take the  $P$ -envelopes to obtain  $\eta_j$  and  $\psi_j$ . It remains to compare them with the limit. Fix  $j$ . For every  $k \geq j$ ,

$$\eta_j \leq \eta_k \leq u_k.$$

Apply the closure of the  $P$ -order under  $d_S$ -convergence [Xia26a, Proposition 6.2.5] to the constant sequence  $\eta_j$  and the tail  $(u_k)_{k \geq j}$ . Their model envelopes satisfy  $P_\theta[\eta_j] \leq P_\theta[u]$ . Since both potentials are model, this is  $\eta_j \leq u$ . Likewise,  $u_k \leq \psi_k \leq \psi_j$  for  $k \geq j$ ; order closure applied to the tail  $(u_k)_{k \geq j}$  and the constant sequence  $\psi_j$  gives  $P_\theta[u] \leq P_\theta[\psi_j]$ , hence  $u \leq \psi_j$ . Together with  $\eta_j \leq u_j \leq \psi_j$ , this proves the assertion.  $\square$

**2.2. Mixed volumes and movable intersections.** We retain the mixed-volume normalization fixed in Theorem 1.2. Mixed volume is symmetric, multilinear for Minkowski addition, monotone, translation invariant, and continuous for the Hausdorff topology; see [Sch14, Section 5.1].

For big numerical divisor classes  $\alpha_1, \dots, \alpha_n$  on a smooth projective  $X$ , we write  $\langle \alpha_1, \dots, \alpha_n \rangle$  for their movable intersection number. The nef-approximation description is [BFJ09, Definition 2.5 and Proposition 2.13]. If  $\mu: Y \rightarrow X$  is a modification with  $Y$  smooth and

$$\mu^* \alpha_i = \beta_i + \{E_i\}, \quad \beta_i \text{ nef}, \quad E_i \text{ is an effective } \mathbb{R}\text{-divisor},$$

then

$$(2.6) \quad \beta_1 \cdots \beta_n \leq \langle \alpha_1, \dots, \alpha_n \rangle.$$

For nef  $\alpha_i$ , the movable and ordinary intersection numbers agree by [BFJ09, Proposition 2.12].

### 3. AN ALGEBRAIC UPPER BOUND

**Lemma 3.1.** *Let  $F/\mathbb{C}$  be a finitely generated field of transcendence degree  $n$ , let  $X$  be an integral projective model with  $\mathbb{C}(X) = F$ , and let  $\nu$  be a surjective valuation which is trivial on  $\mathbb{C}^\times$ , written as*

$$\nu: F^\times \longrightarrow \mathbb{Z}_{\text{lex}}^n.$$

*Let  $W_1, \dots, W_n \subset F$  be non-zero finite-dimensional vector spaces. Evaluation of rational functions defines*

$$\Phi_i: X \dashrightarrow \mathbb{P}(W_i^\vee), \quad x \longmapsto [\text{ev}_{i,x}], \quad \text{ev}_{i,x}(f) := f(x).$$

*This map is defined wherever all elements of  $W_i$  are regular and  $\text{ev}_{i,x} \neq 0$ . Put*

$$\Phi := (\Phi_1, \dots, \Phi_n): X \dashrightarrow \prod_{i=1}^n \mathbb{P}(W_i^\vee),$$

*and let  $Y$  be the Zariski closure of its image. Assume that  $\Phi$  is birational onto  $Y$  and that*

$$\Lambda := \text{span}_{\mathbb{Z}}\{\nu(f) - \nu(g) : f, g \in W_i^\times, 1 \leq i \leq n\} = \mathbb{Z}^n.$$

*Put  $P_i := \text{Conv } \nu(W_i^\times)$ . If  $\pi_i$  is the projection of the ambient product onto  $\mathbb{P}(W_i^\vee)$ , put*

$$H_i := c_1(\pi_i^* \mathcal{O}_{\mathbb{P}(W_i^\vee)}(1)).$$

*Then*

$$(3.1) \quad n! \text{vol}(P_1, \dots, P_n) \leq (H_1 \cdots H_n \cdot Y).$$

*Proof.* If  $\dim_{\mathbb{C}} W_i = 1$  for some  $i$ , then  $P_i$  is a point and  $H_i = 0$ , since the  $i$ -th factor is  $\mathbb{P}^0$ . Both sides of (3.1) are therefore zero. We may henceforth assume that  $\dim_{\mathbb{C}} W_i \geq 2$  for every  $i$ ; in particular, the general hyperplanes used below exist in every factor.

For  $a \in \mathbb{Z}^n$ , let

$$\mathcal{F}_{\succeq a} := \{0\} \cup \{f \in F^\times : \nu(f) \succeq a\}, \quad \mathcal{F}_{\succ a} := \{0\} \cup \{f \in F^\times : \nu(f) \succ a\},$$

where  $\succeq$  is the lexicographic order. The associated graded ring and the initial form of  $f \in F^\times$  are

$$\mathrm{gr}_\nu F := \bigoplus_{a \in \mathbb{Z}^n} \mathcal{F}_{\succeq a} / \mathcal{F}_{\succ a}, \quad \mathrm{in}_\nu(f) := f \bmod \mathcal{F}_{\succ \nu(f)}.$$

The quotient  $\mathcal{F}_{\succeq a} / \mathcal{F}_{\succ a}$  is called the leaf of  $\nu$  at  $a$ . Since  $\nu$  has rational rank  $n$ , Abhyankar's inequality forces its residue field to be  $\mathbb{C}$ ; hence every non-zero graded piece is one-dimensional over  $\mathbb{C}$ . Indeed, if two non-zero elements have the same value, the residue of their quotient is a scalar, and subtracting the corresponding scalar multiple raises the value. Equivalently, every non-zero finite-dimensional subspace  $V \subset F$  satisfies

$$\#\nu(V^\times) = \dim_{\mathbb{C}} V;$$

compare [LM09, Lemma 1.3] and [Xia26a, Section 10.3].

Put  $r_i := \dim_{\mathbb{C}} W_i - 1$ . Choose one  $f_{ij} \in W_i^\times$  above each value in  $\nu(W_i^\times)$  and set

$$a_{ij} := \nu(f_{ij}), \quad 0 \leq j \leq r_i.$$

The  $a_{ij}$  are pairwise distinct, so the valuation axiom makes the  $f_{ij}$  linearly independent. They therefore form a basis of  $W_i$ , and

$$\nu(W_i^\times) = \{a_{i0}, \dots, a_{ir_i}\}.$$

*The initial degeneration.* Choose  $u_\ell \in F^\times$  with  $\nu(u_\ell) = e_\ell$ . One-dimensionality of the leaves gives a non-canonical graded isomorphism. More precisely, for  $a = (a_1, \dots, a_n) \in \mathbb{Z}^n$ , write  $u^a := u_1^{a_1} \cdots u_n^{a_n}$ , and write

$$\mathbb{C}[\mathbb{Z}^n] := \bigoplus_{a \in \mathbb{Z}^n} \mathbb{C}\chi^a, \quad \chi^a \chi^{a'} = \chi^{a+a'},$$

for the group algebra of  $\mathbb{Z}^n$ . Then

$$\mathbb{C}[\mathbb{Z}^n] \xrightarrow{\sim} \mathrm{gr}_\nu F, \quad \chi^a \mapsto \mathrm{in}_\nu(u^a).$$

After rescaling the  $f_{ij}$ , which merely changes the homogeneous coordinates, we may and do assume that

$$\mathrm{in}_\nu(f_{ij}) = \chi^{a_{ij}}.$$

Let  $x_{ij}$  be the homogeneous coordinate on  $\mathbb{P}(W_i^\vee)$  dual to  $f_{ij}$ . In these coordinates,

$$\Phi_i(x) = [f_{i0}(x) : \cdots : f_{ir_i}(x)].$$

Let  $\epsilon_1, \dots, \epsilon_n$  be the standard basis of the multidegree lattice  $\mathbb{Z}^n$ . The multihomogeneous coordinate ring of the ambient product is

$$S := \mathbb{C}[x_{ij} : 1 \leq i \leq n, 0 \leq j \leq r_i], \quad \deg x_{ij} := \epsilon_i.$$

Let  $I \subseteq S$  be the multihomogeneous prime ideal of  $Y$ . Equivalently, a multihomogeneous polynomial  $G$  belongs to  $I$  exactly when  $G(f_{ij}) = 0$  in  $F$ .

Recall that each  $a_{ij}$  is a column vector in  $\mathbb{Z}^n$ . Put  $N := \sum_{i=1}^n (r_i + 1)$ , and define  $A$  to be the  $n \times N$  matrix whose column indexed by  $(i, j)$  is  $a_{ij}$ . Equivalently,

$$A := [a_{10} \cdots a_{1r_1} \mid \cdots \mid a_{n0} \cdots a_{nr_n}] \in \mathbb{Z}^{n \times N}.$$

For  $\alpha = (\alpha_{ij}) \in \mathbb{N}^N$ , write  $x^\alpha := \prod_{i,j} x_{ij}^{\alpha_{ij}}$ , and define

$$\mathrm{wt}_A(x^\alpha) := \sum_{i,j} \alpha_{ij} a_{ij} \in \mathbb{Z}^n.$$

These weights are compared lexicographically. If  $0 \neq G = \sum_{\alpha} \gamma_{\alpha} x^{\alpha}$ , let  $\text{in}_{\mathbf{A}}(G)$  be the sum of the terms of least  $\mathbf{A}$ -weight, and put

$$Q := \text{in}_{\mathbf{A}}(I) := (\text{in}_{\mathbf{A}}(G) : 0 \neq G \in I).$$

Let  $t_1, \dots, t_n$  be algebraically independent variables of multidegrees  $\deg t_i = \epsilon_i$ , and define

$$J := \ker(S \longrightarrow \mathbb{C}[\mathbb{Z}^n][t_1, \dots, t_n], \quad x_{ij} \longmapsto \chi^{a_{ij}} t_i).$$

If  $G \in I$  is multihomogeneous, then  $G(f_{ij}) = 0$ . Taking initial forms of the terms of least valuation in this equality gives  $\text{in}_{\mathbf{A}}(G) \in J$ . Since  $I$  is multihomogeneous, the initial forms of its multihomogeneous elements generate  $Q$ ; hence

$$Q \subseteq J.$$

This is the argument of [KM19, Lemma 2.16].

By [KM19, Proposition 8.10], there is a rational weight vector  $\omega = (\omega_{ij})$  such that  $Q = \text{in}_{\omega}(I)$ , where  $\text{in}_{\omega}$  is defined by taking terms of least  $\omega$ -weight. After clearing denominators, the  $\omega_{ij}$  may be taken integral. The weight filtration on  $S/I$  respects the  $\mathbb{N}^n$ -grading and satisfies

$$\text{gr}_{\omega}(S/I) \simeq S/\text{in}_{\omega}(I) = S/Q.$$

Each multihomogeneous piece is finite-dimensional, so passing to its associated graded preserves dimension. Hence

$$(3.2) \quad \dim_{\mathbb{C}}(S/I)_{\mathbf{d}} = \dim_{\mathbb{C}}(S/Q)_{\mathbf{d}} \quad (\mathbf{d} \in \mathbb{N}^n).$$

*The toric component.* Let  $T := (\mathbb{C}^{\times})^n$ . For  $z = (z_1, \dots, z_n) \in T$  and  $a = (a_1, \dots, a_n) \in \mathbb{Z}^n$ , write  $z^a := z_1^{a_1} \cdots z_n^{a_n}$ , and define

$$\tau: T \longrightarrow \prod_{i=1}^n \mathbb{P}(W_i^{\vee}), \quad z \longmapsto ([z^{a_{i0}} : \dots : z^{a_{ir_i}}])_{i=1}^n.$$

If  $G \in S$  is multihomogeneous of multidegree  $\mathbf{d} = (d_1, \dots, d_n)$ , then its image under the homomorphism defining  $J$  is

$$t_1^{d_1} \cdots t_n^{d_n} G(\chi^{a_{ij}}).$$

Thus  $J$  is precisely the multihomogeneous ideal of the closure of  $\tau(T)$ , and

$$Z := \text{MultiProj}(S/J) = \overline{\tau(T)}.$$

On the dense torus of the ambient product,  $\tau$  is defined by the characters  $z^{a_{ij} - a_{i0}}$ . Their exponents generate  $\Lambda = \mathbb{Z}^n$ , so the induced homomorphism of character rings is surjective. Hence  $\tau$  is a closed immersion into that torus. We identify  $T$  with its image, a dense open subset of  $Z$ ; in particular,  $Z$  is irreducible of dimension  $n$ .

Put  $Y_0 := \text{MultiProj}(S/Q)$ . The inclusion  $Q \subseteq J$  gives a closed immersion  $Z \hookrightarrow Y_0$ . The ideal  $Q$  need not be saturated with respect to the multigraded irrelevant ideal, but this causes no change in the argument: passing to  $\text{MultiProj}$  removes precisely the irrelevant torsion, and the multigraded Serre theorem identifies the sufficiently positive graded pieces with those of the corresponding sheafification. Thus (3.2) gives a common multigraded Hilbert polynomial for  $Y$  and  $Y_0$ . Its restriction to the diagonal is the Hilbert polynomial of the ample line bundle  $\mathcal{O}(1, \dots, 1)$ ; hence  $\dim Y_0 = \dim Y = n$ . Consequently,  $Z$  is a top-dimensional irreducible component of  $Y_0$ .

For an  $n$ -dimensional closed subscheme  $V$  of the ambient product, the part of total degree  $n$  of its multigraded Hilbert polynomial is

$$\frac{1}{n!} \left( \left( \sum_{i=1}^n d_i H_i \right)^n \cdot [V]_n \right),$$

where  $[V]_n$  is its  $n$ -dimensional fundamental cycle. Comparing the coefficient of  $d_1 \cdots d_n$  yields

$$(H_1 \cdots H_n \cdot Y) = (H_1 \cdots H_n \cdot [Y_0]_n).$$

Write

$$[Y_0]_n = \sum_W m_W [W], \quad m_W := \ell(\mathcal{O}_{Y_0, \eta_W}) \geq 1.$$

Here  $W$  runs over the  $n$ -dimensional irreducible components of  $Y_0$  and  $\eta_W$  is the generic point of  $W$ . In particular,  $Z$  occurs with coefficient  $m_Z \geq 1$ . Since the  $H_i$  are globally generated, their mixed degree on every component is non-negative. Therefore

$$(3.3) \quad (H_1 \cdots H_n \cdot Y) \geq m_Z (H_1 \cdots H_n \cdot Z) \geq (H_1 \cdots H_n \cdot Z).$$

*The mixed degree of the toric component.* The  $i$ -th exponent polytope of  $\tau$  is  $P_i$ . General hyperplanes pull back to general Laurent polynomials with Newton polytopes  $P_1, \dots, P_n$ . Since their systems are basepoint-free and  $\dim(Z \setminus T) \leq n - 1$ , an incidence count shows that their common zero set avoids the boundary. The Bernstein–Kushnirenko theorem, or equivalently the toric mixed-intersection formula [Ber75; Ful93], gives

$$(H_1 \cdots H_n \cdot Z) = n! \operatorname{vol}(P_1, \dots, P_n).$$

Together with (3.3), this proves (3.1).  $\square$

**Proposition 3.2.** *Let  $X$  be a smooth connected projective variety of dimension  $n$ , let  $D_1, \dots, D_n$  be big  $\mathbb{Q}$ -divisors, and let  $\nu: \mathbb{C}(X)^\times \rightarrow \mathbb{Z}_{\text{lex}}^n$  be surjective and trivial on  $\mathbb{C}^\times$ . Then*

$$n! \operatorname{vol}(\Delta_\nu(D_1), \dots, \Delta_\nu(D_n)) \leq \langle D_1, \dots, D_n \rangle.$$

Here  $\Delta_\nu(D)$  for a big  $\mathbb{Q}$ -divisor is defined by homogeneity after clearing denominators.

*Proof.* By homogeneity, assume that all  $D_i$  are integral. Let  $x_\nu$  be the center of  $\nu$  on  $X$ . For each  $i$ , choose a non-zero rational section  $\tau_i$  of  $\mathcal{O}_X(D_i)$  which is regular and nowhere vanishing near  $x_\nu$ . For every integer  $k \geq 1$ , put

$$W_{i,k} := H^0(X, \mathcal{O}_X(kD_i)) \tau_i^{-k} \subset \mathbb{C}(X).$$

Whenever  $W_{i,k} \neq 0$ , put

$$P_{i,k} := \operatorname{Conv} \nu(W_{i,k}^\times).$$

Let  $\Phi_k$  be the joint evaluation map associated with the  $W_{i,k}$ , as in Lemma 3.1:

$$\Phi_k = (\Phi_{1,k}, \dots, \Phi_{n,k}): X \dashrightarrow \prod_{i=1}^n \mathbb{P}(W_{i,k}^\vee).$$

Under multiplication by  $\tau_i^k$ , its  $i$ -th component is the Kodaira map of  $|kD_i|$ .

The proof of [Xia26a, Proposition 10.3.1] supplies an integer  $p_0 > 0$  and  $s_0, s_1, \dots, s_n \in W_{1,p_0}^\times$  with

$$\nu(s_j) - \nu(s_0) = e_j, \quad 1 \leq j \leq n.$$

Choose a sufficiently divisible multiple  $p = rp_0$  such that every map defined by  $|pD_i|$  is birational. Set

$$\tilde{s}_0 := s_0^r, \quad \tilde{s}_j := s_j s_0^{r-1} \quad (1 \leq j \leq n).$$

These elements lie in  $W_{1,p}^\times$  and satisfy  $\nu(\tilde{s}_j) - \nu(\tilde{s}_0) = e_j$ . If  $k = mp$ , then  $\tilde{s}_0^m, \tilde{s}_j \tilde{s}_0^{m-1} \in W_{1,k}^\times$ , so

$$\operatorname{span}_{\mathbb{Z}} \{ \nu(f) - \nu(g) : f, g \in W_{i,k}^\times, 1 \leq i \leq n \} = \mathbb{Z}^n.$$

Moreover, the image of the multiplication map

$$\operatorname{Sym}^m W_{i,p} \longrightarrow W_{i,k}$$

is a linear subsystem defining the  $m$ -th Veronese of the birational map associated with  $|pD_i|$ . Hence  $\Phi_{i,k}$ , and therefore  $\Phi_k$ , is birational onto its image.

Let  $Y_k$  be the closure of the image of  $\Phi_k$ , and let  $H_{i,k}$  be the restriction to  $Y_k$  of the hyperplane class from the  $i$ -th factor. Let  $\mu_k: X_k \rightarrow X$  be a common resolution of the base ideals of the systems  $|kD_i|$ , with  $X_k$  smooth, and write

$$\mu_k^*(kD_i) = M_{i,k} + F_{i,k}, \quad F_{i,k} \geq 0,$$



where  $M_{i,k}$  is basepoint-free and  $F_{i,k}$  is the effective fixed part. The induced morphism  $X_k \rightarrow Y_k$  is birational and pulls  $H_{i,k}$  back to  $M_{i,k}$ . Hence the projection formula and [Lemma 3.1](#) give

$$\begin{aligned} n! \operatorname{vol}(P_{1,k}, \dots, P_{n,k}) &\leq (H_{1,k} \cdots H_{n,k} \cdot Y_k) \\ &= M_{1,k} \cdots M_{n,k} \\ &\leq k^n \langle D_1, \dots, D_n \rangle. \end{aligned}$$

The last inequality is [\(2.6\)](#), since  $\mu_k^* D_i - k^{-1} M_{i,k} = k^{-1} F_{i,k} \geq 0$ .

Finally, [\[Xia26a, Proposition 10.3.4\]](#) gives

$$k^{-1} P_{i,k} = \Delta_{\nu,k}(D_i) \xrightarrow{d_{\text{Haus}}} \Delta_{\nu}(D_i).$$

This holds in particular along  $k \in p\mathbb{Z}_{>0}$ . Divide the preceding inequality by  $k^n$  and use continuity of mixed volume.  $\square$

#### 4. UNIFORMITY IN THE VALUATION

We first isolate a convex-geometric compactness statement. It is important that neither the position nor the shape of the bodies is fixed. Recall that  $\mathcal{K}_n$  is the set of convex bodies in  $\mathbb{R}^n$ .

**Lemma 4.1.** *For  $j \in \mathbb{Z}_{>0}$  and  $1 \leq i \leq n$ , let  $A_{i,j}, C_{i,j}, D_{i,j}, B_{i,j} \in \mathcal{K}_n$  satisfy*

$$A_{i,j} \subseteq C_{i,j} \subseteq B_{i,j}, \quad A_{i,j} \subseteq D_{i,j} \subseteq B_{i,j}.$$

*Suppose that there are constants  $v, M > 0$ , independent of  $i$  and  $j$ , such that*

$$\begin{aligned} \operatorname{vol}(A_{i,j}) &\geq v, \\ \operatorname{vol}(B_{1,j} + \cdots + B_{n,j}) &\leq M, \\ \operatorname{vol}(B_{i,j}) - \operatorname{vol}(A_{i,j}) &\longrightarrow 0 \quad (1 \leq i \leq n). \end{aligned}$$

*Then*

$$|\operatorname{vol}(C_{1,j}, \dots, C_{n,j}) - \operatorname{vol}(D_{1,j}, \dots, D_{n,j})| \longrightarrow 0.$$

We refer to [\[Sch14\]](#) for John's theorem and Blaschke selection theorem used in the proof below.

*Proof.* Translate each  $i$ -th quadruple by the same vector so that  $0 \in A_{i,j}$ , and set  $S_j := B_{1,j} + \cdots + B_{n,j}$ . These translations do not change mixed volumes. Every body is now contained in  $S_j$ , and

$$v \leq \operatorname{vol}(S_j) \leq M.$$

Let  $\mathbb{B}^n \subseteq \mathbb{R}^n$  be the closed Euclidean unit ball. Choose an affine isomorphism  $\mathcal{T}_j(x) := L_j x + b_j$ , with  $L_j \in \operatorname{GL}_n(\mathbb{R})$  and  $b_j \in \mathbb{R}^n$ , which sends the John ellipsoid of  $S_j$  to  $\mathbb{B}^n$ . Writing  $\kappa_n := \operatorname{vol}(\mathbb{B}^n)$ , John's theorem gives

$$\mathbb{B}^n \subseteq \mathcal{T}_j(S_j) \subseteq n\mathbb{B}^n, \quad \frac{\kappa_n}{M} \leq |\det L_j| \leq \frac{n^n \kappa_n}{v}.$$

Thus the transformed bodies lie in one fixed compact set, the transformed  $A_{i,j}$  have volume bounded away from zero, and their transformed volume gaps tend to zero.

Starting with any subsequence, Blaschke's selection theorem gives a further subsequence along which all  $4n$  transformed bodies converge. At the limit,

$$\hat{A}_i \subseteq \hat{C}_i \subseteq \hat{B}_i, \quad \hat{A}_i \subseteq \hat{D}_i \subseteq \hat{B}_i, \quad \operatorname{vol}(\hat{A}_i) = \operatorname{vol}(\hat{B}_i) > 0,$$

where the hats denote the respective Hausdorff limits. Hence  $\hat{A}_i = \hat{C}_i = \hat{D}_i = \hat{B}_i$ . Continuity of mixed volume now makes the transformed mixed-volume difference tend to zero. Finally,

$$\operatorname{vol}(\mathcal{T}_j(K_1), \dots, \mathcal{T}_j(K_n)) = |\det L_j| \operatorname{vol}(K_1, \dots, K_n)$$

for arbitrary  $K_1, \dots, K_n \in \mathcal{K}_n$ ; the affine translations are harmless. The lower bound for  $|\det L_j|$  gives the same conclusion before transformation. Since this works for every subsequence, the full sequence converges.  $\square$



The next result upgrades fixed-valuation continuity only at the level of mixed-volume values. It does not assert valuation-uniform Hausdorff convergence of the bodies themselves.

**Proposition 4.2.** *For  $1 \leq i \leq n$ , fix a big line bundle  $L_i$ , a smooth reference metric with curvature  $\theta_i$ , and potentials  $\varphi_{i,j}, \varphi_i \in \text{PSH}(X, \theta_i)_{>0}$  such that*

$$\varphi_{i,j} \xrightarrow{d_S} \varphi_i.$$

Then

$$(4.1) \quad \sup_{\nu} \left| \text{vol}(\Delta_{\nu}(\theta_1, \varphi_{1,j}), \dots, \Delta_{\nu}(\theta_n, \varphi_{n,j})) - \text{vol}(\Delta_{\nu}(\theta_1, \varphi_1), \dots, \Delta_{\nu}(\theta_n, \varphi_n)) \right| \longrightarrow 0,$$

where the supremum is over all surjective valuations  $\mathbb{C}(X)^{\times} \rightarrow \mathbb{Z}_{\text{lex}}^n$  that are trivial on  $\mathbb{C}^{\times}$ .

*Proof.* Replace every  $\varphi_{i,j}$  and  $\varphi_i$  by its model envelope. As recalled in [Section 2](#), this changes neither its  $d_S$ -class nor its partial Okounkov body.

Suppose that (4.1) fails. Then there are a subsequence, still indexed by  $j$ , valuations  $\nu_j$ , and a constant  $\delta > 0$  for which the absolute difference in (4.1), evaluated at  $\nu_j$ , is at least  $\delta$ .

Apply [Lemma 2.1](#) successively for  $i = 1, \dots, n$ , at each step restricting the subsequence obtained at the preceding step. After this finite diagonal extraction, there are increasing model potentials  $\eta_{i,j}$  and decreasing model potentials  $\psi_{i,j}$  on one common subsequence such that

$$(4.2) \quad \eta_{i,j} \leq \varphi_{i,j} \leq \psi_{i,j}, \quad \eta_{i,j} \leq \varphi_i \leq \psi_{i,j}, \quad \eta_{i,j} \xrightarrow{d_S} \varphi_i, \quad \psi_{i,j} \xrightarrow{d_S} \varphi_i.$$

By the  $d_S$ -continuity of raw non-pluripolar mass [[Xia26a](#), Corollary 6.2.5], the diagonal masses of  $\eta_{i,j}$  and  $\psi_{i,j}$  converge to the positive raw mass of  $\varphi_i$ . We may therefore discard one finite initial segment so that all partial Okounkov bodies occurring below lie on the positive-volume locus.

For every  $i, j$ , put

$$\begin{aligned} A_{i,j} &:= \Delta_{\nu_j}(\theta_i, \eta_{i,j}), & B_{i,j} &:= \Delta_{\nu_j}(\theta_i, \psi_{i,j}), \\ C_{i,j} &:= \Delta_{\nu_j}(\theta_i, \varphi_{i,j}), & D_{i,j} &:= \Delta_{\nu_j}(\theta_i, \varphi_i). \end{aligned}$$

By (2.3) and (4.2),

$$A_{i,j} \subseteq C_{i,j} \subseteq B_{i,j}, \quad A_{i,j} \subseteq D_{i,j} \subseteq B_{i,j}.$$

The numerical bounds in [Lemma 4.1](#) do not depend on  $\nu_j$ . Indeed, (2.2) and the  $d_S$ -continuity of the volume [[Xia26a](#), Corollary 7.3.1] give

$$\begin{aligned} \text{vol}(B_{i,j}) - \text{vol}(A_{i,j}) &= \frac{1}{n!} \left( \text{vol}(\theta_i + \text{dd}^c \psi_{i,j}) - \text{vol}(\theta_i + \text{dd}^c \eta_{i,j}) \right) \longrightarrow 0, \\ \text{vol}(A_{i,j}) &= \frac{1}{n!} \text{vol}(\theta_i + \text{dd}^c \eta_{i,j}) \longrightarrow \frac{1}{n!} \text{vol}(\theta_i + \text{dd}^c \varphi_i) > 0. \end{aligned}$$

Furthermore, (2.4) and (2.3) yield

$$B_{1,j} + \dots + B_{n,j} \subseteq \Delta_{\nu_j} \left( \sum_i \theta_i + \text{dd}^c \psi_{i,j} \right) \subseteq \Delta_{\nu_j}(L_1 \otimes \dots \otimes L_n).$$

By (2.5), the last body has Euclidean volume  $\text{vol}(L_1 \otimes \dots \otimes L_n)/n!$ , independently of  $j$  and  $\nu_j$ .

All assumptions of [Lemma 4.1](#) are therefore satisfied. It forces the mixed-volume difference at  $\nu_j$  to tend to zero, contradicting its lower bound  $\delta$ .  $\square$

## 5. PROOF OF THE MIXED-VOLUME FORMULA

**Proposition 5.1.** *The conclusion of [Theorem 1.2](#) holds if each  $h_i$  has analytic singularities and every  $c_1(L_i, h_i)$  is a Kähler current.*

*Proof.* Take a projective log resolution  $\pi: Y \rightarrow X$ , with  $Y$  smooth and projective, for which

$$\pi^* c_1(L_i, h_i) = [D_i] + R_i,$$

where  $D_i$  is an effective  $\mathbb{Q}$ -divisor and  $R_i$  is a closed positive current with locally bounded potentials; see [Xia26a, Definition 1.6.2 and Theorem 1.6.1]. Put

$$\alpha_i := \{R_i\} = \pi^* c_1(L_i) - \{D_i\} \in N^1(Y)_{\mathbb{Q}}.$$

The class  $\alpha_i$  is nef by [Xia26a, Lemma 1.6.1]. Choose a nef  $\mathbb{Q}$ -Cartier divisor in this numerical class and denote it again by  $\alpha_i$ . This is harmless because ordinary Okounkov bodies depend only on numerical classes for the present valuations [Xia26a, Proposition 10.3.2]. Then

$$(5.1) \quad \int_X c_1(L_1, h_1) \wedge \cdots \wedge c_1(L_n, h_n) = \int_Y R_1 \wedge \cdots \wedge R_n = \alpha_1 \cap \cdots \cap \alpha_n.$$

Here we used [Xia26a, Proposition 2.4.2 and Proposition 2.2.1(1), (3), and (4)]. Note that

$$\alpha_i^n = \int_X c_1(L_i, h_i)^n > 0,$$

so the nef divisor  $\alpha_i$  is big.

For a  $\mathbb{Q}$ -Cartier divisor  $D$  and a valuation  $\nu$  centered at  $y_\nu \in Y$ , choose  $m > 0$  such that  $mD$  is Cartier and let  $g$  be a local equation of  $mD$  at  $y_\nu$ . Set

$$\nu(D) := m^{-1} \nu(g) \in \mathbb{Q}^n.$$

For every valuation  $\nu$ , the log-resolution formula [Xia26a, Remark 10.3.1, Equation (10.22)] reads

$$(5.2) \quad \Delta_\nu(L_i, h_i) = \Delta_\nu(\alpha_i) + \nu(D_i).$$

Translation invariance, Proposition 3.2, and nefness give

$$(5.3) \quad n! \operatorname{vol}(\Delta_\nu(L_1, h_1), \dots, \Delta_\nu(L_n, h_n)) \leq \langle \alpha_1, \dots, \alpha_n \rangle = \alpha_1 \cap \cdots \cap \alpha_n.$$

It remains to prove the opposite inequality after taking the supremum. Fix an ample  $\mathbb{Q}$ -divisor  $H$  on  $Y$ . Choose one  $c \in \mathbb{Q}_{>0}$  so large that every  $c\alpha_i - H$  is big, and choose effective  $\mathbb{Q}$ -divisors

$$E_i \sim_{\mathbb{Q}} c\alpha_i - H.$$

For  $\epsilon \in \mathbb{Q}_{>0}$  set

$$\beta_i := \alpha_i + \epsilon H.$$

The  $\mathbb{Q}$ -divisor  $\beta_i$  is ample and

$$(5.4) \quad \frac{1}{1+c\epsilon} \beta_i + \frac{\epsilon}{1+c\epsilon} E_i \sim_{\mathbb{Q}} \alpha_i.$$

By [Wil25, Theorem 1.1], there is an admissible flag

$$Y_\bullet: Y = Y_0 \supset Y_1 \supset \cdots \supset Y_n = \{y\},$$

where each  $Y_r$  is irreducible of codimension  $r$  in  $Y$  and is nonsingular at  $y$ . Let  $\nu_\epsilon$  be the valuation defined by this flag. It satisfies

$$(5.5) \quad n! \operatorname{vol}(\Delta_{\nu_\epsilon}(\beta_1), \dots, \Delta_{\nu_\epsilon}(\beta_n)) = \beta_1 \cap \cdots \cap \beta_n.$$

The flag valuation is surjective and trivial on  $\mathbb{C}^\times$ ; since  $\mathbb{C}(Y) = \mathbb{C}(X)$ , it is among the valuations in Theorem 1.2. After clearing denominators, multiplication by the canonical section of  $E_i$  gives

$$\Delta_{\nu_\epsilon}(\beta_i) + \epsilon \nu_\epsilon(E_i) \subseteq \Delta_{\nu_\epsilon}(\beta_i + \epsilon E_i).$$

By homogeneity, numerical invariance [Xia26a, Proposition 10.3.2], and (5.4), this becomes

$$\frac{1}{1+c\epsilon} \Delta_{\nu_\epsilon}(\beta_i) + \frac{\epsilon}{1+c\epsilon} \nu_\epsilon(E_i) \subseteq \Delta_{\nu_\epsilon}(\alpha_i).$$

Using (5.5), we find

$$(5.6) \quad \sup_{\nu} \operatorname{vol}(\Delta_{\nu}(\alpha_1), \dots, \Delta_{\nu}(\alpha_n)) \geq \frac{1}{n!(1+c\epsilon)^n} \beta_1 \cdots \beta_n.$$

Letting  $\epsilon \rightarrow 0+$  in (5.6), and then using the translations in (5.2), proves the reverse of (5.3). Together with (5.1), this proves the proposition.  $\square$

*Proof of Theorem 1.2.* Choose smooth Hermitian metrics  $h_i^0$  on  $L_i$  and write

$$\theta_i := c_1(L_i, h_i^0), \quad h_i = h_i^0 e^{-\varphi_i}.$$

Set

$$\tilde{\varphi}_i := P_{\theta_i}[\varphi_i]_{\mathcal{I}}.$$

By [Xia26a, Proposition 3.2.6],  $\varphi_i$  and  $\tilde{\varphi}_i$  are  $\mathcal{I}$ -equivalent, so their partial Okounkov bodies agree for every  $\nu$ . By the definition of Cao's mixed volume, the left-hand side of (1.4) is the mixed non-pluripolar mass of the  $\tilde{\varphi}_i$ . Moreover,

$$\int_X \theta_{i, \tilde{\varphi}_i}^n = \text{vol}(\theta_i + \text{dd}^c \varphi_i) > 0,$$

and each  $\tilde{\varphi}_i$  is an  $\mathcal{I}$ -good model potential by [Xia26a, Example 7.1.2]. We may therefore replace the metrics by  $h_i^0 e^{-\tilde{\varphi}_i}$  and relabel  $\tilde{\varphi}_i$  as  $\varphi_i$ ; this changes neither side of (1.4) and places every potential in  $\text{PSH}(X, \theta_i)_{>0}$ .

[Xia26a, Theorem 7.1.1] gives potentials  $\psi_{i,j}$  with analytic singularities such that

$$\psi_{i,j} \xrightarrow{ds} \varphi_i, \quad \theta_{i, \psi_{i,j}} \text{ is a Kähler current.}$$

By Proposition 5.1, for every  $j$ ,

$$(5.7) \quad \int_X \theta_{1, \psi_{1,j}} \wedge \cdots \wedge \theta_{n, \psi_{n,j}} = n! \sup_{\nu} \text{vol}(\Delta_{\nu}(\theta_1, \psi_{1,j}), \dots, \Delta_{\nu}(\theta_n, \psi_{n,j})).$$

By [Xia26a, Theorem 6.2.1], the left-hand side converges to

$$\int_X c_1(L_1, h_1) \wedge \cdots \wedge c_1(L_n, h_n).$$

By Proposition 4.2, the right-hand side converges to

$$n! \sup_{\nu} \text{vol}(\Delta_{\nu}(L_1, h_1), \dots, \Delta_{\nu}(L_n, h_n)).$$

Indeed, if  $f_j(\nu)$  and  $f(\nu)$  denote the corresponding mixed-volume values, then

$$\left| \sup_{\nu} f_j(\nu) - \sup_{\nu} f(\nu) \right| \leq \sup_{\nu} |f_j(\nu) - f(\nu)| \rightarrow 0$$

by Eq. (4.1). Taking limits in (5.7) proves (1.4).  $\square$

*Proof of Theorem 1.1.* For each  $i$ , choose a psh metric  $h_i$  with minimal singularities on  $L_i$ . The partial body is contained in the ordinary body,

$$\Delta_{\nu}(L_i, h_i) \subseteq \Delta_{\nu}(L_i).$$

The diagonal formulas (1.3) and (1.1), together with  $\text{vol dd}^c h_i = \text{vol}(L_i)$ , show that these two compact convex sets have the same positive Euclidean volume. Strict volume monotonicity for full-dimensional convex bodies therefore gives

$$\Delta_{\nu}(L_i, h_i) = \Delta_{\nu}(L_i)$$

for every valuation  $\nu$  under consideration. On the other hand, [BEGZ10, Definition 1.17] identifies the mixed non-pluripolar product of currents with minimal singularities with the positive intersection product; in the projective numerical setting this is the movable product of [BFJ09, Propositions 2.12–2.13]. Applying Theorem 1.2 to the metrics  $h_i$  now turns its left-hand side into  $\langle L_1, \dots, L_n \rangle$  and its partial bodies into the ordinary bodies. The resulting equality is (1.2).  $\square$

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